

BUS33 - 3

Corporate Finance

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About this textbook

This year, for the first time, every Bachelor course at Albert School has its own textbook, thanks to your teachers and AI.

- The textbook is a help for working on the course at home, after the class.
- The course as taught in class remains the reference, and prevails over the textbook wherever the two differ.
- This is a first version. Help us improve it by sharing your feedback with your teacher.

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Preface

Before this course you learned to **read** and **diagnose** a company: to treat the three financial statements as constructed documents, to compute the ratios and working capital that summarise a firm's health, and to isolate operating performance with **ROIC** and the **free cash flow** that valuation discounts. You also met the central test of value creation — a firm creates value only when $\text{ROIC} > \text{WACC}$ — but left the **WACC**, the cost of capital, as a black box. This course opens that box.

Here you stop describing companies and start **deciding for** them. The subject organises around Aswath Damodaran's three corporate-finance decisions:

- **Invest** — where to deploy capital. Which projects earn more than they cost?
- **Finance** — how to fund it. What mix of debt and equity, and at what cost?
- **Return** — how much to give back. Dividends, buybacks, or reinvestment?

Every tool here serves one of the three. Time value of money, NPV and IRR answer **invest**. The cost of equity, the cost of debt and WACC answer **finance** and supply the discount rate that **invest** needs. Capital structure and payout policy complete **finance** and **return**. And the value-creation metrics — ROIC against WACC, and EVA — deliver the verdict on whether the three decisions, taken together, made the firm worth more.

We keep one running company throughout, **Lumio**, a mid-size lighting manufacturer, so that every new tool refers back to one set of numbers rather than a fresh example each time (Table 1). Lumio's operations earn a return on invested capital of about 10.5% — by the end of the book you will know the **WACC** that number must beat, and what to do with the surplus it creates.

Item (€m)		Item (€m)	
Revenue	600	Net financial debt	220
EBIT	80	financial debt	260
Depreciation & amortisation	40	cash	40
EBITDA	120	Book equity	350
Net income	45	Invested capital	570
NOPAT (= 80×0.75)	60	Interest expense	20

Table 1: Lumio's headline figures, referred to throughout (effective tax rate 25%). From these, $\text{NOPAT} = \text{EBIT} \times (1 - \tau) = 60$ and $\text{ROIC} = \text{NOPAT} / \text{invested capital} = 60/570 \approx 10.5\%$.

1. Time value of money

Example — why €1,000 next year is not €1,000 today. A supplier offers Lumio a choice: €1,000 now, or €1,000 in one year. The answer is obvious — take it now — but **why**? Because €1,000 today can be placed at, say, 5% and become €1,050 in a year, so €1,000 in a year is worth **less** than €1,000 today. Exactly how much less? Whatever sum, placed at 5% today, grows to €1,000 in a year: $€1,000/1.05 \approx €952$. That €952 is what next year's €1,000 is **worth now**. Every valuation in this book is a longer version of this one division.

A sum of money available today is not equal to the same nominal sum available later: money today can be invested to earn a return, so it is worth more. The **time value of money** is the machinery that makes cash flows at different dates comparable — and comparison is the first step of every decision that follows.

Definition 1 (Future and present value). At a periodic rate r , a sum C invested for n periods grows by **compound interest** — interest earning interest — to its **future value**

$$FV = C(1 + r)^n.$$

Conversely, the **present value** of a sum C_n received n periods from now is found by **discounting**:

$$PV = \frac{C_n}{(1 + r)^n}.$$

The factor $1/(1 + r)^n$ is the **discount factor**; the rate r used to discount is the **discount rate**.

Example — compounding forward, discounting back. Place €2,000 at 6% for three years. It grows to $FV = 2000 \times 1.06^3 \approx €2,382$. Now reverse the question: a payment of €2,382 due in three years, discounted at 6%, is worth $2382/1.06^3 \approx €2,000$ today. Compounding and discounting are the **same operation run in opposite directions** — one multiplies a sum forward, the other divides it back.

The further away a cash flow, and the higher the rate, the more discounting shrinks it — which is why distant, risky cash flows contribute so little to value today.

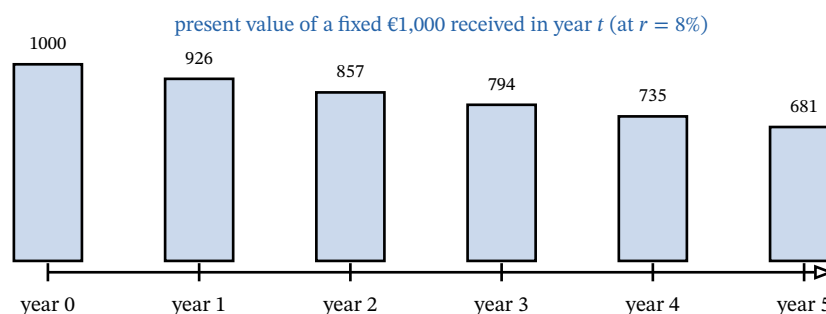


Figure 1: Discounting at work. A flat €1,000 promised in each future year is worth progressively less today: €1,000 now, but only €681 if it arrives in five years. The discount factor $1/(1 + r)^t$ erodes value geometrically with distance and with the rate.

For a **stream** of cash flows $C_1, C_2, \dots, C_n$ at the end of periods 1 through n , present values simply add — each is carried back to today on its own and then summed:

$$PV = \sum_{t=1}^n \frac{C_t}{(1+r)^t}.$$

The same stream valued at a **future** date — say the end of period n — is found by carrying each flow **forward** instead of back, then adding:

$$FV_n = \sum_{t=1}^n C_t(1+r)^{n-t}.$$

Example — future value of a stream: what regular deposits grow to. Lumio sets aside €10,000 at the end of each of the next three years, earning 5%. By the end of year 3 the first deposit has compounded for two years, the second for one, the third not at all:

$$FV_3 = 10000 \times 1.05^2 + 10000 \times 1.05 + 10000 = \text{€}31,525.$$

Present value carries a stream back to today; future value carries it forward to a common later date — the same machinery pointed either way.

Some streams run so long that summing term by term is hopeless — a maintenance contract paying every year for a decade, or a stable cash flow assumed to continue forever. Three such streams recur often enough to have closed forms that collapse the whole sum into one division.

Proposition 1 (Perpetuities and annuities).

- A **perpetuity** paying C at the end of every period forever is worth

$$PV = \frac{C}{r}.$$

- A **growing perpetuity** whose payment grows at rate $g < r$ each period (first payment C) is worth

$$PV = \frac{C}{r-g}.$$

The plain perpetuity is the special case $g = 0$.

- An **annuity** paying C for n periods is worth

$$PV = \frac{C}{r} \left(1 - \frac{1}{(1+r)^n} \right).$$

Example — an annuity: a fixed stream with an end date. A maintenance contract pays Lumio €50,000 at the end of each year for 8 years. At a 6% discount rate its value today is

$$PV = \frac{50000}{0.06} \left(1 - \frac{1}{1.06^8} \right) \approx 50000 \times 6.21 \approx \text{€}310,000.$$

Summing eight discounted terms by hand would give the same figure; the annuity formula does it in one line.

Example — the growing perpetuity is the engine of valuation. Suppose Lumio's free cash flow next year is €30 m and is expected to grow at $g = 2\%$ forever, discounted at $r = 8\%$. Its value today is

$$PV = \frac{30}{0.08 - 0.02} = \frac{30}{0.06} = €500 \text{ m.}$$

This single formula — the **Gordon growth** model — is the terminal-value engine behind almost every company valuation. Notice how sensitive it is: raise g to 3% and the value jumps to €600 m. Small assumptions about the far future drive large fractions of value, a fragility we return to in Chapter 10.

Pitfall. The discount rate must **match** the cash flows in three ways, or the arithmetic is meaningless:

- **Periodicity.** A monthly cash flow is discounted at a monthly rate. An annual rate of 12% is **not** 1% per month compounded — $(1.01)^{12} \approx 12.7\%$, not 12%.
- **Inflation basis.** Discount **nominal** cash flows (which include expected inflation) at a **nominal** rate, and **real** cash flows at a **real** rate. Mixing the two is a classic and large error.
- **Risk.** The rate must reflect the **risk** of these particular cash flows — the subject of Chapter 4, Chapter 5 and Chapter 6. Risk-free cash flows are discounted at the risk-free rate; risky ones demand more.

Everything that follows — NPV, the cost of capital, the whole of valuation — is an application of this single principle: a cash flow is worth its discounted value, and values at the same date may be compared and added.

Exercises

- 1.1. At 4% a year, how much is €5,000 worth after 3 years?
- 1.2. You deposit €2,000 at the end of each year for 4 years into an account paying 4%. How much is in the account just after the fourth deposit?
- 1.3. A payment of €10,000 arrives in 5 years. At a 6% discount rate, what is it worth today?
- 1.4. A project pays €1,000, €2,000 and €1,500 at the end of years 1, 2 and 3. At 7%, what is the present value of the stream?
- 1.5. What is the present value of €4,000 received at the end of each year for 10 years, discounted at 5%?
- 1.6. An asset is expected to pay €500 next year, growing 3% a year forever. At an 8% discount rate, what is it worth today? What if growth were 4% instead?

Answers. (1) $5000 \times 1.04^3 \approx €5,624.32$.

(2) $2000(1.04^3 + 1.04^2 + 1.04 + 1) \approx €8,492.93$ (the future value of a four-payment stream).

(3) $10000/1.06^5 \approx €7,472.58$.

(4) $1000/1.07 + 2000/1.07^2 + 1500/1.07^3 \approx €3,905.90$.

(5) $4000 \times (1 - 1.05^{-10})/0.05 \approx €30,886.94$.

(6) $500/(0.08 - 0.03) = €10,000$ at $g = 4\%$, $500/0.04 = €12,500$ — a one-point change in growth lifts value by 25%.

2. Net Present Value and the investment decision

We now make the first of the three decisions: **invest**. An investment project is a set of dated cash flows — an outflow now (and perhaps later) in exchange for inflows in the future. The question is whether the future inflows, **valued today**, exceed what they cost.

Example — should Lumio buy the new production line? Lumio can install an automated line for €100 m today, which will generate net cash inflows of €30 m, €35 m, €40 m and €25 m over the next four years (then be scrapped at nil value). Is it worth it? You **cannot** simply add the inflows ($30 + 35 + 40 + 25 = €130 \text{ m} > €100 \text{ m}$) and declare victory — those euros arrive at different dates and are not comparable. Discount each to today at the firm's 8% cost of capital, then compare with the €100 m outlay.

Definition 2 (Net Present Value). The **Net Present Value** of a project with cash flows $C_0, C_1, \dots, C_n$ (with C_0 usually negative — the initial outlay) discounted at rate r is

$$\text{NPV} = \sum_{t=0}^n \frac{C_t}{(1+r)^t} = C_0 + \frac{C_1}{1+r} + \dots + \frac{C_n}{(1+r)^n}.$$

NPV measures the value a project creates **today**, after fully paying investors for the time value of money and the risk of the cash flows through the discount rate. A positive NPV is value the project adds to the firm over and above the cost of the capital it ties up.

Example — the production line, valued. At $r = 8\%$:

$$\begin{aligned} \text{NPV} &= -100 + \frac{30}{1.08} + \frac{35}{1.08^2} + \frac{40}{1.08^3} + \frac{25}{1.08^4} \\ &= -100 + 27.8 + 30.0 + 31.8 + 18.4 = +€8.0 \text{ m} . \end{aligned}$$

Discounted, the inflows are worth €108 m against a €100 m cost: the line creates about €8 m of value today. Accept it. Note how different this is from the naïve €30 m surplus the undiscounted sum suggested — discounting ate most of it, and at a high enough rate would have turned the project negative.

Proposition 2 (The NPV rule).

- For a **single** project: accept if $\text{NPV} > 0$, reject if $\text{NPV} < 0$.
- Among **mutually exclusive** projects (you can pick only one): choose the one with the **highest** NPV, provided it is positive.

The rule is not arbitrary: a positive-NPV project returns more than the cost of the capital it uses, so taking it raises the value of the firm by exactly the NPV. Maximising NPV **is** maximising shareholder value — the objective the rest of the book serves.

$$\Sigma \text{ discounted inflows} = 108 > 100 \text{ outlay} \Rightarrow \text{NPV} = +8$$

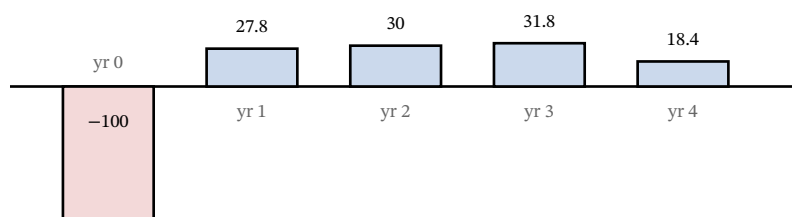


Figure 2: The production line as discounted cash flows (€m). The €100 m outlay at year 0 is set against the **present values** of the four inflows — not their face values. Their sum, €108 m, exceeds the outlay by the €8 m NPV.

The discount rate used is the **cost of capital** (Chapter 4, Chapter 5 and Chapter 6), reflecting the risk of the project's cash flows — for a project of average firm risk, the firm's WACC. A riskier project demands a higher rate, and the same cash flows then create less (or negative) value.

Pitfall. Build NPV on **incremental, after-tax cash flows**, and watch four traps:

- **Sunk costs** are irrelevant — money already spent (a feasibility study) does not change with the decision and must be excluded.
- **Opportunity costs** are relevant — using land Lumio already owns has the cost of what that land could otherwise earn.
- **Cash, not accounting profit** — add back depreciation, subtract capex and working-capital investment, exactly as in the free-cash-flow build you already know.
- **Financing flows belong in the discount rate, not the cash flows** — do not subtract interest from the project's cash flows **and** discount at the WACC; that double-counts the cost of debt.

Exercises

2.1. A project costs €200 m now and returns €80 m at the end of each of the next three years.

Compute its NPV at an 8% cost of capital and give the accept/reject verdict.

2.2. Re-run the same project at a 12% cost of capital. What happens to the verdict, and what does the change tell you about the project's sensitivity to the discount rate?

2.3. Lumio must choose **one** of two mutually exclusive projects at a 10% cost of capital. Project A: outlay €100 m, single inflow €130 m in year 1. Project B: outlay €60 m, single inflow €80 m in year 1. Which should it take, and why is the smaller outlay not automatically the better choice?

Answers. (1) $\text{NPV} = -200 + 80 \times (1 - 1.08^{-3})/0.08 \approx +€6.17 \text{ m} > 0 \rightarrow \text{accept}.$

(2) $\text{NPV} = -200 + 80 \times (1 - 1.12^{-3})/0.12 \approx -€7.85 \text{ m} < 0 \rightarrow \text{reject}.$ The verdict flips between 8% and 12%, so the project's break-even rate lies between them: a thin cushion that a higher hurdle rate erases.

(3) A: $-100 + 130/1.10 \approx +€18.18 \text{ m}.$ B: $-60 + 80/1.10 \approx +€12.73 \text{ m}.$ Take **A**: it adds more value. NPV, not outlay size, ranks mutually exclusive projects.

3. Internal Rate of Return

NPV answers “how much value?” Managers often also want a single **rate of return** they can compare against the cost of capital and quote without restating a discount rate. That number is the IRR.

Example — the rate at which Lumio's line breaks even. Lumio's production line had an NPV of +€8 m at an 8% cost of capital, but we saw a high enough discount rate would wipe that surplus out. At **exactly** what rate does the value vanish? Somewhere between 10% and 12% — about 11.5%. That break-even rate is a figure managers love: “the line returns 11.5%,” set against the 8% the capital costs, needs no discount rate attached to be understood. It is the project's **internal rate of return**.

Definition 3 (Internal Rate of Return). The **Internal Rate of Return** of a project is the discount rate that makes its NPV exactly zero:

$$\sum_{t=0}^n \frac{C_t}{(1 + \text{IRR})^t} = 0.$$

It is the project's own break-even rate — the return it earns on the capital it ties up, expressed as a percentage.

Unlike NPV, the IRR has no closed-form formula for a general cash-flow stream: you find it numerically, by searching for the rate that zeroes the NPV.

Method 1 (Computing the IRR by interpolation). Bracket the IRR with two rates and interpolate between them:

1. Evaluate the NPV at two rates $r_1 < r_2$, chosen so that $\text{NPV}_1 > 0 > \text{NPV}_2$ (one positive, one negative — so the root lies between).
2. Estimate the crossing by **linear interpolation**:

$$\text{IRR} \approx r_1 + (r_2 - r_1) \frac{\text{NPV}_1}{\text{NPV}_1 - \text{NPV}_2}.$$

Narrow the bracket and repeat for more precision. A spreadsheet's IRR function automates exactly this search.

Example — the production line's IRR. For Lumio's line, NPV at 10% is +€3.3 m and at 12% is −€1.0 m, so the root lies between them:

$$\text{IRR} \approx 10\% + 2\% \times \frac{3.3}{3.3 + 1.0} \approx 11.5\%.$$

Read it against the 8% cost of capital: the project earns about 11.5% on its capital, comfortably above the 8% that capital costs, so it creates value — the same verdict NPV gave. The two agree because, for this project, NPV is positive exactly while the discount rate stays below 11.5%.

Proposition 3 (When IRR and NPV agree). For a **conventional** project — one initial outflow followed only by inflows, so the cash flows change sign exactly once — the IRR rule “accept if $IRR > r$ ” gives the **same accept/reject verdict** as the NPV rule. This is because such a project’s NPV falls steadily as the discount rate rises and crosses zero exactly once, at the IRR.

The agreement is only on the **yes/no** decision for a single conventional project. For **ranking** projects, and for unconventional cash flows, IRR can mislead in three distinct ways.

Example — scale: the higher IRR, the lower value. Lumio weighs two mutually exclusive projects at $r = 8\%$:

Project	Outlay	Inflow yr 1	IRR	NPV at 8%
Small	€10 m	€13 m	30%	€2.0 m
Large	€100 m	€118 m	18%	€9.3 m

IRR prefers Small ($30\% > 18\%$); NPV prefers Large ($€9.3 \text{ m} > €2.0 \text{ m}$). IRR is a **rate** and is blind to **size** — 30% of a small base is less money than 18% of a large one. If Lumio can take only one, it should take Large: it adds more value. NPV ranks correctly; IRR does not.

The three ways the IRR rule breaks:

- **Scale.** As above, IRR ignores project size and can rank a small project above a larger, more valuable one.
- **Timing.** Projects whose cash flows arrive on different schedules can rank differently under IRR than under NPV, and the IRR ranking can even **reverse** as the cost of capital changes — the two swap at a **crossover rate** where their NPV profiles intersect.
- **Multiple or no IRR.** A project whose cash flows change sign **more than once** (an outflow, inflows, then a large clean-up outflow) can have several IRRs, or none — leaving “accept if $IRR > r$ ” undefined.

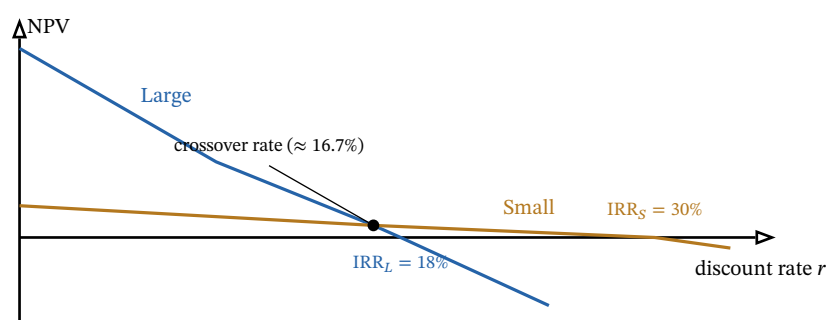


Figure 3: NPV profiles of two mutually exclusive projects. Each curve crosses zero at its own IRR — Small has the **higher** IRR (30%) but the **lower** intercept. The profiles cross at a **crossover rate** of about 16.7%, just left of Large’s 18% zero. At any cost of capital below the crossover, Large has the higher NPV — so the IRR ranking is wrong there. NPV reads off the correct project at the firm’s actual r .

Example — multiple IRRs: when the rule breaks down. A project costs €100 m now, returns €230 m in year 1, then needs a €132 m clean-up outflow in year 2: cash flows $-100, +230, -132$. Setting $x = 1 + IRR$ and solving $-100 + 230/x - 132/x^2 = 0$, i.e. $100x^2 - 230x + 132 = 0$, gives

$x = 1.10$ and $x = 1.20$ — an IRR of **10% and of 20%**. The NPV profile is hump-shaped: negative below 10%, (barely) positive between 10% and 20%, negative again above 20%. At an 8% cost of capital, both roots exceed 8%, so a naïve “accept if $IRR > r$ ” says **accept** — yet the NPV at 8% is about −€0.2 m, so the project should be **rejected**. With two break-even rates the IRR rule has no single answer; NPV settles it directly.

Pitfall. Two persistent errors with IRR:

1. **Treating the higher IRR as the better project.** For mutually exclusive choices, rank by NPV, not IRR — the scale and timing effects above make the IRR ranking unreliable.
2. **Believing the IRR “reinvests interim cash flows at the IRR”.** You will often read that IRR — unlike NPV — silently assumes each cash flow is reinvested at the IRR itself. This is a myth, not a property of the arithmetic. Both $\sum C_t/(1 + IRR)^t = 0$ and $NPV = \sum C_t/(1 + r)^t$ depend only on the project’s own cash flows and a single rate; neither contains any term for what happens to a cash flow **after** it is received. A reinvestment rate enters only if you **choose** to carry the interim flows forward to a terminal value — as the modified IRR (MIRR) does on purpose — which is an added modelling assumption, not something the IRR already does. The real reasons to prefer NPV are the scale and timing effects above, not a reinvestment story.

NPV is the **superior** rule: it measures value added directly, in currency, at the right cost of capital, and never multiplies or vanishes. IRR survives as a **communication** device — a return figure that can be compared against the cost of capital across projects without restating the discount rate — and as an accept/reject test for a single conventional project. Know its failure modes and default to NPV whenever they bite.

Exercises

3.1. A project has cash flows −60, +25, +25, +25 (€m). Estimate its IRR by linear interpolation, using NPV evaluations at 10% and 15%.

3.2. Two mutually exclusive projects: Small has IRR 30% and NPV €2 m at the cost of capital; Large has IRR 18% and NPV €9 m. Which does the IRR rule prefer, which does the NPV rule prefer, and which should the firm take? Name the effect at work.

3.3. A project has cash flows −100, +230, −132 (€m). Show that it has two IRRs, and explain why “accept if $IRR > r$ ” cannot be applied. How would you decide instead?

Answers. (1) $NPV(10\%) = -60 + 25 \times (1 - 1.10^{-3})/0.10 \approx +€2.17$ m; $NPV(15\%) \approx -€2.92$ m. So $IRR \approx 10\% + 5\% \times 2.17/(2.17 + 2.92) \approx 12.1\%$ (exact $\approx 12.0\%$).

(2) IRR prefers Small; NPV prefers Large; the firm should take **Large** — it adds more value. The **scale** effect: IRR is a rate and ignores the size of the investment.

(3) Solving $-100 + 230/x - 132/x^2 = 0$ with $x = 1 + IRR$ gives $x = 1.10$ and $x = 1.20$, so $IRR = 10\%$ and 20% . With two break-even rates the rule is undefined; both exceed an 8% cost of capital, yet NPV at 8% is about −€0.2 m — reject. Decide on NPV directly.

4. Cost of equity via CAPM

The discount rate that NPV and IRR both needed is the **cost of capital** — and to build it we must first price each source of finance separately. We begin with equity. Shareholders are paid last and bear the firm's risk, so they require a return for it. But **which** risk?

Example — the risk that gets paid for. A shareholder who holds only Lumio bears all of Lumio's ups and downs. But a shareholder holding a **diversified portfolio** of hundreds of stocks barely feels a Lumio-specific shock — a factory fire, a product recall — because other holdings offset it. That **firm-specific** (diversifiable) risk can be diversified away for free, so the market does not reward bearing it. What **cannot** be diversified away is the risk Lumio shares with the whole market — recessions, rate shocks. **Only this systematic risk is priced.** This is the insight the CAPM formalises.

Definition 4 (CAPM — the cost of equity). The **Capital Asset Pricing Model** gives the required return on a firm's equity as

$$r_e = r_f + \beta(r_m - r_f),$$

where

- r_f is the **risk-free rate**;
- β (beta) is the stock's sensitivity to market movements — its systematic risk;
- $r_m - r_f$ is the **equity risk premium** (ERP): the extra return investors demand for holding the market portfolio over the risk-free asset.

The product $\beta(r_m - r_f)$ is the stock's own **risk premium**.

Beta is the bridge to your statistics toolkit: it is the covariance of the stock's return with the market's, divided by the variance of the market's return, $\beta = \text{Cov}(r_i, r_m) / \text{Var}(r_m)$ — a standardised measure of how strongly the stock moves with the market. $\beta = 1$ means it moves one-for-one with the market; $\beta > 1$ amplifies market moves (cyclical firms); $\beta < 1$ dampens them (defensive firms such as utilities).

Example — Lumio's cost of equity. Take $r_f = 3\%$ (a long government bond yield), $\beta = 1.2$ (lighting is mildly cyclical), and an equity risk premium of $r_m - r_f = 6\%$. Then

$$r_e = 3\% + 1.2 \times 6\% = 3\% + 7.2\% = 10.2\%.$$

Lumio's shareholders require about 10.2% a year. Notice this sits just **below** its 10.5% ROIC: equity alone — the most expensive source — nearly demands everything Lumio's operations earn. Only once we blend in cheaper after-tax debt (Chapter 6) does a clear value-creation spread open up — a first hint that Lumio's value creation is real but modest.

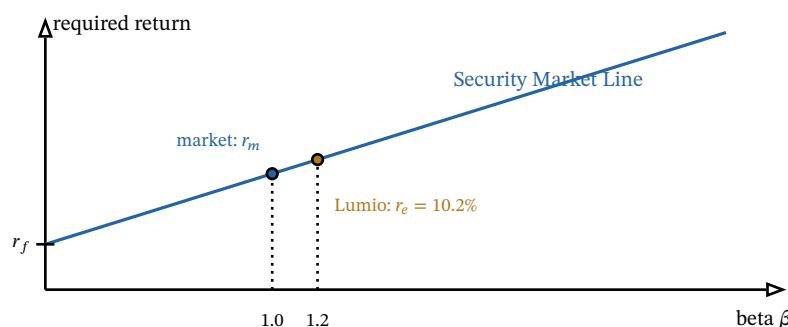


Figure 4: The Security Market Line. Required return rises linearly with systematic risk β : it starts at the risk-free rate ($\beta = 0$), passes through the market return at $\beta = 1$, and places Lumio slightly above the market because its $\beta = 1.2$.

Each input demands judgment — CAPM is a disciplined framework, not a number that falls out of a database:

- **Risk-free rate.** Proxied by a government bond yield, with maturity matched to the horizon of the cash flows being discounted (a long bond for long-lived projects).
- **Beta.** Estimated by regressing the stock's returns on market returns, or sourced from a data provider. It shifts with the estimation window, the return frequency (daily vs monthly), and the index chosen, and is often **adjusted toward 1** because betas tend to revert. For an unlisted firm or division, take a **comparable** firm's beta and re-lever it to the target structure.
- **Equity risk premium.** Estimated either from long-run **historical** average excess returns or **implied** from current market prices; the two can differ by several points, so the choice must be argued, not assumed.

The re-levering the beta bullet calls for, for an unlisted firm or division, has a standard form. A comparable's beta reflects **both** its business risk and the financial risk its own leverage adds, so it must be stripped of that leverage and re-dressed in the target's before use.

Method 2 (Re-levering a comparable's beta). With tax rate τ :

1. **Unlever** the comparable's equity beta β_e to its **asset** beta, removing the comparable's leverage D/E :

$$\beta_u = \frac{\beta_e}{1 + (1 - \tau) \frac{D}{E}}.$$

2. **Re-lever** the asset beta to the target's own leverage $(D/E)^{\text{target}}$:

$$\beta_e^{\text{target}} = \beta_u \left(1 + (1 - \tau) \left(\frac{D}{E} \right)^{\text{target}} \right).$$

Example — beta for an unlisted division. An unlisted lighting division has no traded shares. A listed pure-play comparable has an equity beta of 1.4 at $D/E = 0.8$, and the tax rate is 25%. Unlever it to the business's asset beta,

$$\beta_u = \frac{1.4}{1 + 0.75 \times 0.8} = \frac{1.4}{1.6} = 0.875,$$

then re-lever to the division's target $D/E = 0.5$:

$$\beta_e = 0.875 \times (1 + 0.75 \times 0.5) \approx 1.20.$$

This 1.20 — carrying the comparable's **business** risk but the division's **own** financial risk — is the beta to feed into CAPM.

Pitfall. CAPM prices **systematic** risk only. Do not inflate r_e for risks that diversification removes — a single-product firm's product risk, say — those belong in the **cash-flow** forecast (lower or probability-weighted flows), not in the discount rate. And never use a **single** firm-wide r_e to discount a project of very different risk: a stable utility evaluating a speculative tech venture must use the venture's higher beta, or it will accept value-destroying projects that a single blended rate would wrongly approve.

Exercises

4.1. Estimate a firm's cost of equity with a risk-free rate of 3.5%, a beta of 0.9 and an equity risk premium of 5.5%.

4.2. Recompute the cost of equity if beta is instead 1.3, holding the other inputs. How many percentage points does the estimate move?

4.3. A single-product firm faces a real risk that its one product fails. Should this be handled by raising beta (and so r_e), or somewhere else? Justify in one or two sentences.

Answers. (1) $r_e = 3.5\% + 0.9 \times 5.5\% = 8.45\%$.

(2) $r_e = 3.5\% + 1.3 \times 5.5\% = 10.65\%$ — up 2.2 points. Beta and the ERP are both judgment calls; here beta drove the change, and its estimate (window, frequency, index, adjustment toward 1) is what to defend.

(3) Not in beta. CAPM prices only **systematic** risk; a diversifiable product-specific risk belongs in the **cash-flow** forecast (lower or probability-weighted flows), not the discount rate.

5. Cost of debt

The second source of finance is debt — and it is cheaper than equity for two reasons: lenders are paid **before** shareholders and bear less risk, and interest is **tax-deductible**. Both effects must be quantified.

Example — what Lumio's debt really costs. Lumio's bonds trade at a yield of 5%: that is the return lenders require today, the **pre-tax** cost of debt — not the historical coupon printed on old bonds, which is irrelevant to a decision made now. But 5% overstates the true cost. Because the €1 of interest Lumio pays is deductible, at a 25% tax rate it cuts the tax bill by €0.25. The net cost of that interest is only €0.75 — an **after-tax** rate of $5\% \times (1 - 0.25) = 3.75\%$. Debt is cheap, and the tax system makes it cheaper.

Definition 5 (Pre-tax cost of debt). The **pre-tax cost of debt** r_d is the return a lender requires **today** to hold the firm's debt — the rate at which the firm could borrow now, not the coupon on bonds it issued in the past.

There are two routes to r_d , and a well-run estimate cross-checks one against the other. The first reads it straight off **traded bonds**; the second **builds** it from the risk-free rate and a **credit spread**, for the many firms whose debt is not publicly priced.

Route 1 — yield to maturity. When the firm has traded bonds, its r_d is their **yield to maturity** (YTM): the single discount rate that sets a bond's price equal to the present value of its promised coupons and principal. The YTM is itself a time-value-of-money construction — it is the bond's IRR (Chapter 3).

Example — the YTM route: where Lumio's 5% comes from. Lumio has a bond outstanding: face value €1,000, an annual coupon of €40 (a 4% coupon rate), and three years left to maturity. It trades today at €973 — below par, because its 4% coupon undershoots the return the market now demands. Its YTM is the single rate y that discounts the coupons and principal back to that price:

$$\frac{40}{1+y} + \frac{40}{(1+y)^2} + \frac{1040}{(1+y)^3} = 973.$$

This is exactly the bond's IRR, so solve it by the interpolation of Chapter 3. At $y = 4\%$ the right-hand side is par, €1,000 (so €27 above the price); at $y = 6\%$ it is €946.5 (€26.5 below). The root lies between, and

$$y \approx 4\% + 2\% \times \frac{27}{27 + 26.5} \approx 5.0\%.$$

So the pre-tax cost of debt is about 5% — the figure the running example uses — and, after the 25% tax, an after-tax cost of $5\% \times 0.75 = 3.75\%$. A bond priced **below** par always yields more than its coupon; one **above** par, less.

Route 2 — risk-free rate plus a credit spread. A firm without traded bonds still has a cost of debt: $r_d = r_f + \text{spread}$, where the **credit spread** compensates lenders for default risk and widens as

credit quality falls. You can anchor the spread with a ratio you already compute — **interest coverage** (EBIT / interest) — through a synthetic-rating table.

Interest coverage	Indicative rating	Spread
$\geq 8 \times$	AA or better	$\approx 0.7\%$
$6-8 \times$	A	$\approx 1.1\%$
$4-6 \times$	BBB	$\approx 1.7\%$
$2.5-4 \times$	BB	$\approx 2.8\%$
$1.5-2.5 \times$	B	$\approx 4.5\%$
$< 1.5 \times$	CCC or below	$\approx 8\% +$

Table 2: A synthetic-rating table turns interest coverage into a credit spread. The figures are **illustrative** and drift with the credit cycle; the durable point is the **shape** — spreads widen sharply as coverage thins.

Example — the spread route, cross-checked on Lumio. Suppose Lumio had no traded bond to price. Its interest coverage is $\text{EBIT} / \text{interest} = 80/20 = 4.0 \times$, which Table 2 places at the foot of the BBB band — a spread of roughly 1.5–2%. With a 3% risk-free rate,

$$r_d = r_f + \text{spread} \approx 3\% + 2\% = 5\%,$$

in the same neighbourhood as the 5% its traded bond actually yields — a reassuring cross-check between the two routes.

Definition 6 (After-tax cost of debt and the tax shield). Because interest is tax-deductible, the cost of debt relevant to the firm is the **after-tax cost**:

$$r_d(1 - \tau),$$

where τ is the marginal corporate tax rate. The tax saved because interest is deductible is the **debt tax shield**: on interest expense I , the saving is

$$\tau \times I \text{ per period.}$$

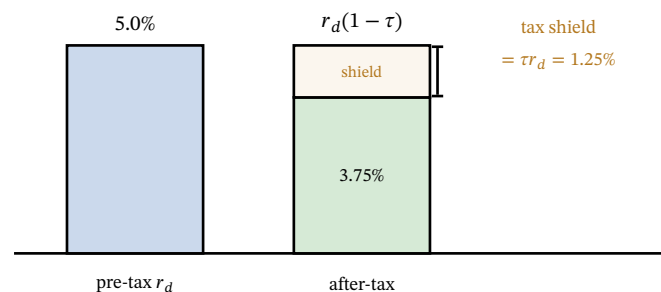


Figure 5: The tax shield on Lumio's debt. A 5% pre-tax cost becomes a 3.75% after-tax cost because the deduction returns $\tau r_d = 0.25 \times 5\% = 1.25\%$ to the firm. The shield is the wedge between the two bars and the mechanism by which leverage can add value (Chapter 7).

The tax shield matters twice over. As a **cost**, it makes debt cheaper than its stated rate — the after-tax rate is what enters WACC next. As a **source of value**, it is the channel through which capital structure can change firm value at all, the hinge of the Modigliani–Miller analysis in Chapter 7.

Pitfall. Use the **current, forward-looking, after-tax** cost of debt for decisions — the rate the firm would pay to borrow **today**. Do not use the average coupon on legacy debt issued years ago at different rates; it is a historical accident, not the cost of new capital. And the shield is only worth τr_d **if the firm is actually paying tax**: a loss-making firm with no taxable profit gets no shield this year, so its after-tax cost of debt is simply its pre-tax cost.

Exercises

- 5.1.** A firm's bonds yield 6%. At a 30% tax rate, what is its after-tax cost of debt?
- 5.2.** A firm has no traded bonds. The risk-free rate is 3% and its credit spread is 2.5%. Estimate its pre-tax cost of debt, then its after-tax cost at a 25% tax rate.
- 5.3.** A firm with no traded bonds covers its interest $3\times$; the risk-free rate is 3%. Use the synthetic-rating table (Table 2) to estimate its pre-tax cost of debt, then its after-tax cost at a 25% tax rate.
- 5.4.** A bond has face value €1,000, pays an annual coupon of €50, and has two years left to maturity. It trades at €1,009. Estimate its yield to maturity — the pre-tax cost of debt — by interpolating between 4% and 5%, then give the after-tax cost at a 25% tax rate.
- 5.5.** A firm pays €12 m of interest a year at a 25% tax rate. What is the annual tax shield? What happens to the shield in a year the firm makes a loss and pays no tax?

Answers. (1) $6\% \times (1 - 0.30) = 4.2\%$.

(2) Pre-tax $r_d = 3\% + 2.5\% = 5.5\%$; after-tax $5.5\% \times 0.75 = 4.125\%$.

(3) Coverage of $3\times$ sits in the BB band, a spread of about 2.8%, so pre-tax $r_d \approx 3\% + 2.8\% = 5.8\%$ and after-tax $5.8\% \times 0.75 \approx 4.35\%$ (the table is illustrative, so read this as a neighbourhood, not a point).

(4) Discounting 50, $50 + 1000$ gives €1,018.9 at 4% (€9.9 above the price) and €1,000.0 at 5% (€9.0 below), so $YTM \approx 4\% + 1\% \times 9.9/(9.9 + 9.0) \approx 4.5\%$; after-tax $4.5\% \times 0.75 \approx 3.4\%$. Above par, the yield sits below the 5% coupon.

(5) Shield = $0.25 \times €12 = €3$ m. In a loss year with no taxable profit there is no tax to save, so the shield is zero and the after-tax cost of debt equals the pre-tax cost.

6. Weighted Average Cost of Capital

A firm financed by both equity and debt must earn enough to satisfy **both** claimant groups. The WACC blends the two costs into the single hurdle rate that the invest decision has been waiting for.

Example — Lumio's blended hurdle rate. Lumio is financed by equity worth $E = €450$ m (market value) and debt worth $D = €260$ m, so $V = 710$ m. Equity costs $r_e = 10.2\%$, debt costs $r_d = 5\%$ pre-tax (3.75% after the 25% tax). The firm's cost of capital is the **weighted average** of the two, by their share of total financing:

$$\text{WACC} = \frac{450}{710} \times 10.2\% + \frac{260}{710} \times 3.75\% = 6.47\% + 1.37\% \approx 7.8\%.$$

This is the roughly-8% hurdle rate the diagnosis stage had to assume. Lumio's ROIC of 10.5% clears it — the firm creates value — but by a spread of under three points, which the rest of the book will learn to defend, widen, or doubt.

Definition 7 (WACC). With market values of equity E and debt D , total $V = E + D$, the **Weighted Average Cost of Capital** is

$$\text{WACC} = \frac{E}{V}r_e + \frac{D}{V}r_d(1 - \tau),$$

where r_e is the cost of equity, r_d the pre-tax cost of debt, and τ the corporate tax rate. It is the minimum return the firm's assets must earn to satisfy all capital providers — and the discount rate for a project of the same risk as the firm as a whole.

The weights are **market-value** (or target) weights, never book weights, because they represent the cost of raising capital **today**: book equity is a historical accounting residual, while the cost of equity is a forward-looking required return that applies to equity's market value.

How does **leverage** move WACC? Replacing expensive equity with cheaper after-tax debt lowers the average — at first. But leverage does not leave the two costs untouched: as debt grows, equity becomes riskier and its required return r_e rises. To see why — and by how much — start from what leverage does to the return equity actually **earns**.

Example — leverage amplifies equity returns — both ways. Take a firm with €100 m of invested capital and an after-tax cost of debt of 4%, and read it in a good year (operating return ROIC = 10%) and a bad one (ROIC = 2%), under two financings:

Financing	ROE, good year	ROE, bad year	swing
All equity ($D/E = 0$)	10%	2%	8 pts
Half debt ($D = E$, $D/E = 1$)	16%	0%	16 pts

With no debt, equity earns exactly what the assets earn. With $D/E = 1$, the good year is lifted to 16% and the bad year sunk to 0%: the swing **doubles**. Leverage magnifies the upside and the downside by the **same** factor, so levered equity is twice as volatile — and its holders demand a higher return for it.

Proposition 4 (Leverage and the return on equity). With after-tax operating return $\text{ROIC} = \text{NOPAT}/\text{invested capital}$ and after-tax cost of debt $r_{d(1-\tau)}$, the return on equity is

$$\text{ROE} = \text{ROIC} + (\text{ROIC} - r_{d(1-\tau)}) \frac{D}{E}.$$

Leverage D/E multiplies the gap between what the assets earn and what the debt costs: when ROIC beats the after-tax cost of debt, leverage lifts ROE above it; when ROIC falls short, leverage drags ROE below — amplifying the downside by exactly the factor it amplifies the upside.

Because levered equity bears this amplified swing, the return its holders **require** must rise with leverage too. Modigliani and Miller pinned down by how much.

Proposition 5 (MM Proposition II — the levered cost of equity). Let r_u be the **unlevered cost of equity** — the return equity would demand if the firm carried no debt (the cost of its assets). With pre-tax cost of debt r_d and tax rate τ , at leverage D/E the cost of equity is

$$r_e = r_u + (r_u - r_d)(1 - \tau) \frac{D}{E}.$$

It climbs linearly with D/E — the required-return counterpart of the ROE amplification above. So re-computing WACC at a new leverage means re-computing r_e from this formula first, never assuming it.

Example — Lumio's WACC at a more leveraged target. What becomes of Lumio's WACC if it moves to a 50/50 target, $D/E = 1$? First recover its unlevered cost of equity from its **current** structure ($D/E = 260/450 \approx 0.58$, $r_e = 10.2\%$, $r_d = 5\%$, $\tau = 25\%$) by reading Proposition II backwards:

$$10.2\% = r_u + (r_u - 5\%)(0.75)(0.58) \implies r_u \approx 8.6\%.$$

Now re-lever to $D/E = 1$ (holding $r_d = 5\%$, still safe at this moderate leverage):

$$r_e = 8.6\% + (8.6\% - 5\%)(0.75)(1) \approx 11.3\%.$$

Equity now costs 11.3%, up from 10.2% — the price of the added risk, **derived**, not guessed. Yet WACC **falls**:

$$\text{WACC} = 0.5 \times 11.3\% + 0.5 \times 5\% \times 0.75 \approx 7.5\%,$$

below the 7.8% at today's structure. Cheaper after-tax debt still outweighs the dearer equity — Lumio sits on the **downward** arm of the WACC curve, consistent with its being mildly under-leveraged (Chapter 7).

Notice what did **not** happen: the rise in r_e did not lift WACC. As long as the cost of debt itself holds steady, it never can. Substitute Proposition II into the WACC definition and the rising cost of equity and equity's shrinking weight cancel **exactly**, leaving

$$\text{WACC} = r_u \left(1 - \tau \frac{D}{V} \right),$$

which slides **straight down** as leverage $\frac{D}{V}$ rises, reaching its lowest at all-debt financing. The extra return levered equity demands is offset euro-for-euro by equity funding a smaller share of the firm, so **on its own it can never turn WACC up**. (Check: with Lumio's $r_u \approx 8.6\%$ and $\tau = 25\%$ this gives 7.8% at today's $D/V \approx 0.37$ and 7.5% at $D/E = 1$ — the two figures we just found the long way.)

What **does** turn WACC back up is the one thing that formula holds fixed: the cost of debt. Push leverage far enough and mounting distress risk **widens the credit spread**, so r_d climbs — and a higher r_d feeds back through Proposition II to push r_e up **further** still. Once debt stops being cheap, both costs rise together and overwhelm the shrinking-weight effect. The result is a **U-shaped WACC curve** — but its upward arm is the work of a **rising cost of debt**, not of levering equity.

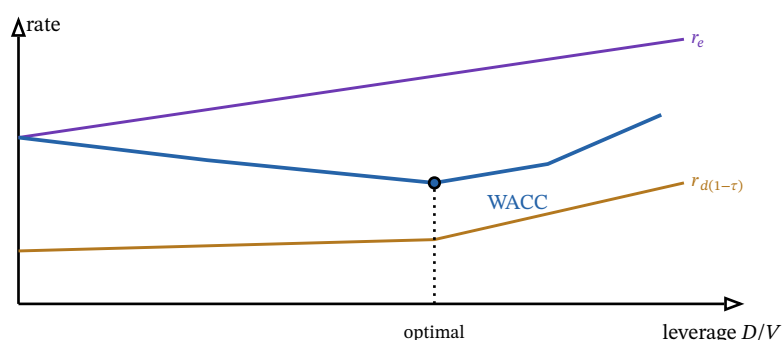


Figure 6: WACC against leverage. While the cost of debt holds steady, replacing equity with cheaper after-tax debt pulls the average **down** — the rising cost of equity is exactly offset by equity's shrinking weight. Only once mounting distress risk **widens the cost of debt** (which then drags equity's up further) does WACC turn back up, giving the curve its U-shape and an interior optimum — the same optimum the trade-off identifies in Chapter 7.

Pitfall. Three WACC errors that quietly corrupt every downstream valuation:

- **Book weights.** Using book equity instead of market equity — especially for a firm whose market value far exceeds book — badly misweights the average.
- **Wrong-risk discounting.** WACC is the right rate only for projects of **firm-average risk**. Discounting a safe cost-saving project and a risky new-market launch at the same WACC over-rewards the risky one and starves the safe one.
- **Double-counting the shield.** The tax benefit of debt is **already** in WACC via $r_{d(1-\tau)}$. Do not **also** add interest tax shields back into the project's cash flows — pick one method (WACC **or** adjusted present value), never both.

Exercises

- 6.1. A firm has €600 m of equity and €400 m of debt (market values), a cost of equity of 12%, a pre-tax cost of debt of 6% and a 25% tax rate. Compute its WACC.
- 6.2. The same firm re-levers to €400 m equity and €600 m debt, its pre-tax cost of debt unchanged at 6%. Using MM Proposition II, first back out its unlevered cost of equity r_u , then derive the new cost of equity, then recompute WACC. Did WACC rise or fall, and would it keep falling if leverage rose without limit?
- 6.3. A firm's book equity is €200 m but its market equity is €800 m. Which figure belongs in the WACC weights, and why?

Answers. (1) $\text{WACC} = 0.6 \times 12\% + 0.4 \times 6\% \times 0.75 = 7.2\% + 1.8\% = 9.0\%$.

(2) Backing out r_u at the original $D/E = 400/600 = 0.667$: $12\% = r_u + (r_u - 6\%)(0.75)(0.667)$, so $r_u = 10\%$. At the

new $D/E = 600/400 = 1.5$, $r_e = 10\% + (10\% - 6\%)(0.75)(1.5) = 14.5\%$, and $\text{WACC} = 0.4 \times 14.5\% + 0.6 \times 6\% \times 0.75 = 5.8\% + 2.7\% = 8.5\%$. It **fell** ($9.0\% \rightarrow 8.5\%$). And with r_d held fixed it **would** keep falling: $\text{WACC} = r_u(1 - \tau \frac{D}{V})$ is strictly decreasing in leverage, bottoming at all-debt financing. What halts the fall in reality is a **rising** r_d — past some leverage, distress risk widens the credit spread, lifting r_d (and, through Proposition II, r_e) enough to turn WACC up: the U-shape.

(3) Market equity (€800 m): WACC is the cost of raising capital **today**, and the cost of equity is a forward-looking return on equity's market value, not its historical book residual.

7. Capital structure

WACC's U-shape already told us that the financing mix can matter. Capital structure theory explains **why**, and starts — paradoxically — by proving that in an idealised world it does **not** matter at all. That benchmark is what makes the real-world frictions legible.

Example — why the pie does not grow by slicing it. Imagine Lumio's operations throw off a fixed stream of cash. Financing it 100% with equity, the shareholders receive the whole stream. Financing it half with debt, lenders take a fixed slice and shareholders take the volatile remainder — but the **two slices still sum to the same stream**. In a world with no taxes and no bankruptcy, an investor could even copy any leverage themselves by borrowing in their own account (“home-made leverage”), so they will not pay a premium for the firm to do it. Value comes from the **assets**, not the financing.

That example is the whole of Modigliani and Miller's first proposition.

Proposition 6 (Modigliani–Miller Proposition I — irrelevance). In a **frictionless** market — no taxes, no bankruptcy costs, no information asymmetry, no transaction costs — the total value of a firm is **independent** of how it is financed. Slicing the same operating cash flows into debt and equity claims does not change the size of the pie:

$$V_{\text{levered}} = V_{\text{unlevered}}.$$

Proposition I concerns the firm's total **value**; its companion, Proposition II (Chapter 6), concerns the **cost of equity**. Together they say leverage only reshuffles risk and return between claimholders — raising the cost of equity in exact step with the risk it loads onto it — without, in this frictionless world, changing the total. MM is a benchmark, not a description of reality: its job is to tell us that **if** capital structure matters, it matters only through the frictions it assumes away.

Relaxing the assumptions one at a time reveals the forces that make structure matter:

- **Taxes (push toward more debt).** Interest deductibility creates the tax shield of Chapter 5. Each euro of permanent debt adds roughly $\tau \times$ that debt in present-value shield, raising firm value as leverage rises.
- **Distress costs (push toward less debt).** Higher leverage raises the probability of financial distress and bankruptcy, which carry **direct** costs (legal and administrative fees) and, larger, **indirect** costs (lost customers and key staff, fire-sale asset disposals, managers absorbed in crisis management instead of investing).
- **Agency costs (cut both ways).** Debt can **discipline** managers by committing free cash flow to interest, curbing wasteful expansion — managers growing the firm beyond its profitable size (a benefit). But heavy debt also breeds **conflicts** between shareholders and creditors — risk-shifting, under-investment — that destroy value (a cost).

Principle 1 (The trade-off theory). Firm value as a function of leverage is the unlevered value **plus** the present value of the tax shield **minus** the present value of expected distress and agency costs:

$$V_{\text{levered}} = V_{\text{unlevered}} + \text{PV}(\text{tax shield}) - \text{PV}(\text{distress costs}).$$

The shield grows roughly linearly with debt; distress costs accelerate as debt rises. Their net defines an interior **optimal capital structure** — the leverage that maximises value (equivalently, minimises WACC).

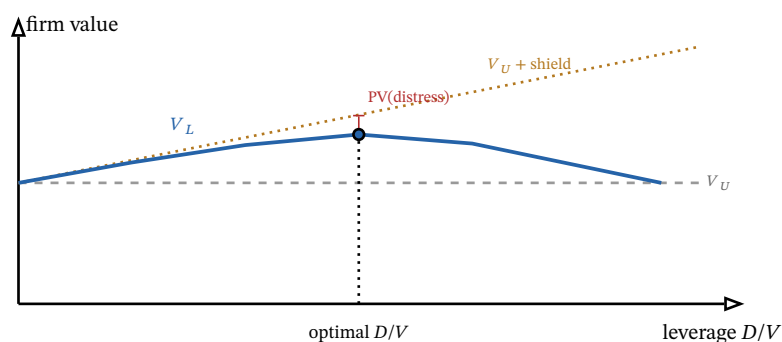


Figure 7: The trade-off theory. The tax shield (dotted) would raise value without limit, but expected distress costs bend the actual value curve (solid) back down — so V_L lies **below** the $V_U + \text{shield}$ line everywhere, the vertical gap being the present value of expected distress costs. Value peaks at an interior optimum: too little debt forgoes the shield, too much invites distress.

Definition 8 (Over- and under-leverage). Relative to its optimal range, a firm is **over-leveraged** when distress and agency costs dominate — thin interest coverage, constrained investment, fragile to shocks — and **under-leveraged** when it leaves an unused tax shield on the table and forgoes debt’s disciplining effect, often holding more idle cash than it needs.

Diagnosing which means locating the firm against its target using the **leverage and coverage ratios** you already compute — debt-to-equity, net-debt-to-EBITDA, interest coverage — read alongside its **business risk**.

Example — is Lumio over- or under-leveraged?. Lumio’s interest coverage is $80/20 = 4.0 \times$ and, with EBITDA of €120 m (EBIT €80 m plus €40 m depreciation and amortisation), its net-debt-to-EBITDA is $220/120 \approx 1.8 \times$ — comfortable for a stable manufacturer. It is, if anything, **mildly under-leveraged**: a manufacturer with steady cash flows could likely carry more debt, capture more tax shield, and lower its WACC below 7.8% without serious distress risk. A volatile tech firm with the **same** ratios, however, might be at its limit — the safe debt level depends on **business risk**, not the ratios alone.

Pitfall. The optimum is a **range**, not a knife-edge, and it is **industry- and risk-specific**. A stable utility safely carries leverage that would bankrupt a cyclical, asset-light software firm. Never benchmark a company’s leverage against the whole market — only against peers of similar business risk and asset tangibility (asset-heavy firms borrow more because their assets back the debt). And remember the shield is worthless to a firm that pays no tax: do not prescribe debt to a chronic loss-maker.

Exercises

7.1. State two assumptions of the Modigliani–Miller irrelevance proposition, and for each name the real-world friction that its removal introduces.

7.2. A stable manufacturer has interest coverage of $6\times$ and net-debt-to-EBITDA of $1.2\times$, against a sector norm of $2.5\times$. Is it more likely over- or under-leveraged, and what single piece of information would most change your answer?

7.3. Explain why prescribing “more debt to capture the tax shield” can be wrong for a firm that has been loss-making for several years.

Answers. **(1)** e.g. **no taxes** (removing it introduces the debt tax shield, favouring leverage) and **no bankruptcy costs** (removing it introduces distress costs, discouraging leverage). Also acceptable: no agency costs, no information asymmetry.

(2) Under-leveraged: strong coverage and leverage below the sector norm leave unused tax shield. Its **business risk** (cash-flow volatility) would most change the answer — the same ratios are safe for a stable firm and dangerous for a volatile one.

(3) The shield is worth $\tau \times$ interest only if there is taxable profit to shelter; a chronic loss-maker has none, so extra debt adds distress risk without the offsetting shield.

8. Payout policy

The third decision — **return** — arises once a firm generates cash beyond its profitable investment needs. How much should go back to shareholders, and through which mechanism? The prior question is always **retention vs distribution**.

Example — Lumio's surplus has to go somewhere. Say Lumio throws off €70 m of cash this year, but its positive-NPV projects — the only ones earning above the cost of capital — absorb just €50 m. The remaining €20 m has no home inside the firm that beats the WACC. Retaining it would earn shareholders the WACC at best, which they could earn themselves by investing elsewhere; so the €20 m should go **back** to them. That residual is what payout policy is about.

Principle 2 (The retention-vs-distribution rule). Cash should be **retained and reinvested** only in projects that earn more than the cost of capital — i.e. positive-NPV projects, equivalently those with $ROIC > WACC$. Cash with no such home should be **distributed**: retaining it to earn the WACC (or less) inside the firm creates no value the shareholder could not create alone. Payout policy is what to do with the residual after all positive-NPV investment is funded.

This connects the three decisions into one logic: **invest** claims every positive-NPV euro; **finance** sets the WACC that defines “positive”; **return** distributes the rest. A firm distributing cash while passing up positive-NPV projects, or holding idle cash with no projects to fund, has its policy backwards.

Definition 9 (Payout mechanisms).

- A **dividend** is a cash distribution per share, typically paid on a regular schedule.
- A **share buyback** (repurchase) returns cash by the firm buying its own shares in the market, reducing the share count — so each remaining share owns a larger slice of the firm.

In a frictionless MM world the **choice** is irrelevant: a shareholder wanting cash can sell shares to make a “homemade dividend,” and a euro paid out is a euro the share price drops.

Example — dividend or buyback: identical in a frictionless world. A firm's equity is worth €500 m, in 50 m shares — €10.00 a share — and it will return €50 m of surplus cash. Compare the two mechanisms for a holder of 100 shares (worth €1,000 today):

	Pay €50 m dividend	Buy back €50 m
Cash returned	€1.00 per share	buys 5 m shares
Shares remaining	50 m	45 m
Value per share after	$(500 - 50)/50 = €9.00$	$(500 - 50)/45 = €10.00$
Holder's 100 shares	€900 + €100 cash	€1,000 (sells none)
Holder's total wealth	€1,000	€1,000

Either way the holder ends with €1,000. Under the buyback, a holder who **wants** cash sells 10 shares for €100 and keeps 90 worth €900 — a “homemade dividend” matching what the dividend would have paid. The euro paid out is exactly the euro the per-share value (dividend) or the share count (buyback) adjusts by; nothing is created or destroyed. And EPS? If the firm earns €40 m,

EPS rises from $40/50 = \text{€}0.80$ to $40/45 \approx \text{€}0.89$ after the buyback — up 11% — though each share is worth exactly what it was. That gap between higher EPS and unchanged value is the trap the Pitfall below returns to.

The choice starts to matter once two frictions enter:

- **Taxes.** Dividends and the capital gains realised through buybacks are often taxed **differently**, and at different times — buybacks defer the tax until the investor sells, dividends tax on receipt. Which mechanism is tax-favoured depends on jurisdiction and shareholder type.
- **Signalling.** Because cutting an established dividend is painful and punished by the market, **raising** the regular dividend is a **credible signal** of confidence in durable future cash flows — managers will not commit unless they believe it. Buybacks are more **flexible** and **discretionary**, so they signal less commitment but suit lumpy or uncertain surplus cash; they also avoid locking the firm into a payout it may later have to cut.

Example — two firms, two right answers. A mature, cash-rich Lumio with stable cash flows and few high-return projects suits a **steady, growing dividend**: it can credibly commit, and its income-oriented shareholders value the regularity. A young, volatile growth firm with lumpy surplus cash and a tax-sensitive, capital-gains-oriented shareholder base suits **buybacks**: flexible, tax-deferring, and not a promise it might have to break. Same surplus, opposite mechanisms — driven by growth stage, cash-flow stability, and the shareholder base.

Pitfall. Buybacks do **not** mechanically “create value” by raising earnings per share. EPS rises only because the share count falls; if the shares are repurchased **above** their intrinsic value, the buyback **destroys** value for continuing shareholders even as EPS ticks up. Judge a buyback like any investment — by price versus value, not by its effect on EPS. And a dividend funded by **new borrowing** rather than free cash flow is a warning sign, not a sign of health.

Exercises

- 8.1.** In two or three sentences, contrast dividends and buybacks on (a) tax timing and (b) the signal each sends.
- 8.2.** Recommend a payout mechanism for each, justified by signalling and tax rather than a payout-ratio rule of thumb: (a) a mature, cash-rich firm with stable flows and income-oriented shareholders; (b) a young firm with volatile, lumpy surplus cash and tax-sensitive shareholders.
- 8.3.** A CEO argues a buyback “creates value because it raises EPS.” State the flaw and the correct test for whether a buyback is worthwhile.

Answers. (1) Dividends are taxed on receipt; the capital gain from a buyback is deferred until the investor sells. Raising a regular dividend is a strong, hard-to-reverse signal of confidence; a buyback is flexible and signals less commitment. (2) (a) A steady, modestly growing **dividend**: the firm can credibly commit and its shareholders value regularity. (b) **Buybacks**: flexible, tax-deferring, and not a promise that lumpy cash might force it to break. (3) EPS rises only because the share count falls. If shares are bought **above** intrinsic value the buyback destroys value for continuing shareholders despite higher EPS. Judge it by price versus value, like any investment.

9. Value-creation metrics

We can now close the loop left open earlier. A firm creates value only when it earns more on its invested capital than that capital costs — and we finally have both halves of the comparison: ROIC (which you already know how to compute) and WACC (Chapter 6).

Example — Lumio's verdict. Lumio earns $\text{ROIC} = 10.5\%$ on its €570 m of invested capital, while that capital costs $\text{WACC} \approx 7.8\%$. Each euro invested therefore earns a $10.5\% - 7.8\% = 2.7\%$ surplus over its cost, so the firm creates about

$$2.7\% \times 570 \approx \text{€}15 \text{ m of value a year.}$$

Equivalently, Lumio's €60 m of after-tax operating profit (NOPAT) is **charged** $7.8\% \times 570 \approx \text{€}44.5$ m for the capital that produced it, leaving $60 - 44.5 \approx \text{€}15.5$ m of genuine economic value (the two routes agree up to rounding). A firm reporting the same €45 m net income but earning $\text{ROIC} < \text{WACC}$ would be **destroying** value despite its positive bottom line — the distinction accounting profit cannot make.

The comparison and the currency figure it implies each have a name.

Definition 10 (The ROIC-vs-WACC test). With $\text{ROIC} = \text{NOPAT} / \text{invested capital}$ and WACC the cost of capital (Chapter 6), the firm

- **creates** value when $\text{ROIC} > \text{WACC}$,
- is **value-neutral** when $\text{ROIC} = \text{WACC}$,
- **destroys** value when $\text{ROIC} < \text{WACC}$.

The gap $\text{ROIC} - \text{WACC}$ is the **economic spread**.

Definition 11 (Economic Value Added). **Economic Value Added (EVA)** expresses the same test as a **currency** amount per period:

$$\text{EVA} = (\text{ROIC} - \text{WACC}) \times \text{invested capital} = \text{NOPAT} - \text{WACC} \times \text{invested capital}.$$

The term $\text{WACC} \times \text{invested capital}$ is the **capital charge** — the euro cost of all the capital tied up.

Positive EVA is value created over the period; negative EVA is value destroyed, however healthy the accounting profit looks.

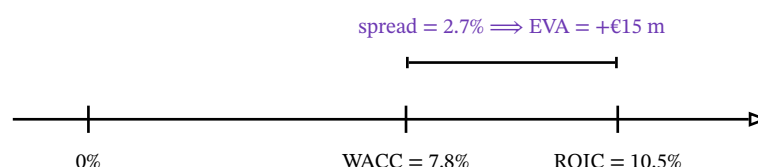


Figure 8: Value creation is a **comparison**, not a level. Lumio's 10.5% ROIC clears its 7.8% WACC, so each euro of the €570 m invested capital earns a 2.7% surplus — about €15 m of EVA a year. The same ROIC against a 12% WACC would destroy value despite looking healthy in isolation.

EVA charges the firm for **all** capital employed — including equity, which accounting profit treats as free — at its true cost. This is precisely why a company can report positive net income yet negative EVA: net income subtracts the cost of debt (interest) but never the cost of equity. Computing ROIC and EVA turns the cost of capital from Chapter 4, Chapter 5 and Chapter 6 into a verdict on whether the company was **worth operating**.

Pitfall. EVA inherits every measurement choice in ROIC and WACC. Use **NOPAT** (operating profit after tax, before interest), not net income, and keep non-operating cash out of invested capital. On the WACC side, a small error in beta or the ERP can swing the capital charge by millions and flip the EVA sign — so a borderline EVA verdict should be reported as a **range**, not a false-precision point estimate.

Exercises

- 9.1.** A firm has NOPAT €90 m, invested capital €750 m and a WACC of 9%. Compute its ROIC and its EVA, and state whether it creates or destroys value.
- 9.2.** Another firm reports the same €90 m NOPAT but has invested capital of €1,200 m at the same 9% WACC. Compute its EVA. What does the comparison show about reading net income alone?
- 9.3.** Return to the first firm. If a re-estimate of beta pushes WACC from 9% to 11%, recompute EVA. What does this say about reporting a borderline EVA as a point estimate?

Answers. (1) $\text{ROIC} = 90/750 = 12\%$; $\text{spread} = 12\% - 9\% = 3\%$; $\text{EVA} = 3\% \times 750 = \text{€}22.5 \text{ m}$ (or $90 - 0.09 \times 750$). Value **created**.

(2) $\text{EVA} = 90 - 0.09 \times 1200 = -\text{€}18 \text{ m}$. Same accounting profit, opposite verdict: net income ignores the cost of the equity capital tied up, which EVA charges.

(3) $\text{EVA} = 90 - 0.11 \times 750 = \text{€}7.5 \text{ m}$ — still positive but a third of before. A two-point WACC change swings the verdict, so a borderline EVA should be reported as a range, not a false-precision point.

10. The integrated investment case

Everything converges here. The three decisions — **invest**, **finance**, **return** — are not taken in isolation but **together**, for one company, using the accounting-adjusted inputs you already know how to build (normalised earnings, free cash flow, ROIC). This is the deliverable the whole sequence has been building toward, and the foundation a full company valuation later assumes you can produce.

Method 3 (The integrated case). For a single company, assemble the three decisions on **one consistent set of assumptions**:

1. **Invest.** Build the project or strategy as dated, incremental, after-tax **free cash flows** and evaluate it with **NPV** at the appropriate discount rate, cross-checked with **IRR** (Chapter 2, Chapter 3).
2. **Finance.** Choose a financing mix; compute its **WACC** (Chapter 6); assess how the proposed leverage moves the cost of capital and where it sits against the firm's **optimal structure** (Chapter 7). Feed this WACC back as the discount rate in step 1.
3. **Return.** Justify a **payout policy** for the cash not absorbed by positive-NPV investment, consistent with the firm's growth, cash-flow stability, and shareholder base (Chapter 8).

The case is coherent only if the **same** inputs, risk assumptions, and tax rate run through all three — the WACC that discounts the project must be the WACC implied by the financing mix, and the cash distributed must be what is left after funding every positive-NPV project.

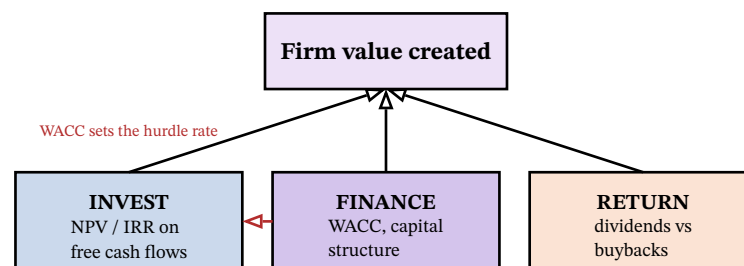


Figure 9: The three corporate-finance decisions feed one objective — value created — and interlock. Finance supplies the WACC that Invest discounts at (red arrow); Invest claims every positive-NPV euro; Return distributes only the residual. They also **compete** for one limited pool of cash, so a sound case resolves the tension between them rather than reading three verdicts that happen to agree.

Example — Lumio's integrated case, where the decisions collide. Read at low stakes, the three decisions agree. The €100 m production line has NPV +€8 m at the 7.8% WACC and an IRR of 11.5% — **invest**, accept. Lumio is mildly under-leveraged at $D/V \approx 37\%$, so a little new debt would capture more tax shield and nudge WACC down, **strengthening** that NPV — **finance** helps invest. And the surplus left over funds the steady dividend its income-oriented base values — **return** follows cleanly. If the story stopped here it would teach nothing: every decision is obvious.

Now make the stakes real. This year Lumio's **whole slate** of positive-NPV projects — the line plus others — needs about €200 m, but internal free cash flow is only €70 m, and its shareholders rely on a €30 m dividend. The three decisions now pull against one another, and no choice satisfies all three:

- **Invest** wants the full €200 m — every project clears the hurdle.

- **Finance** can raise the missing €130 m as debt, but that lifts net debt from €220 m to €350 m, or net-debt-to-EBITDA from $1.8\times$ to nearly $3\times$; protecting the dividend on top pushes it past $3\times$. At that leverage Lumio swings from **under-** to **over-**leveraged: distress risk climbs, the cost of equity re-levers upward (Proposition II, Chapter 6, puts numbers on the rise) as D/E jumps and r_d widens too, so WACC starts going **up** — eroding the very NPVs the debt was meant to fund.
- **Return** could close the gap by cutting the dividend, but that breaks a credible commitment to an income base and signals distress — a real cost, not a free source of cash.

Integrating is **resolving** this, not stacking three yes-votes. A defensible answer ranks the slate by NPV and funds the top projects with about €80 m of new debt — holding net-debt-to-EBITDA near the $\approx 2.5\times$ optimum (net debt €300 m) and interest coverage above $3\times$ — **defers** the marginal projects to next year rather than buying WACC-destroying leverage, and holds the dividend **flat** (neither cut nor grown) for a year. Something gives: a few positive-NPV euros wait. That is the point — the trade-off is unavoidable, and the skill is pricing which decision to bend.

10.1. Critiquing a capital allocation proposal

A finished case is not a verdict to be trusted but an argument to be **stress-tested**. The most valuable skill in corporate finance is knowing **where a model is weakest** — because a positive NPV is only as good as the assumptions underneath it.

Framework 1 (A disciplined critique). Interrogate any capital-allocation proposal along three axes:

1. **Which assumptions drive the NPV?** Identify the few inputs the conclusion is most sensitive to — typically the **growth rate**, **margins**, **terminal value**, and **discount rate** — by varying each and watching the NPV move (sensitivity and scenario analysis). A project whose positive NPV survives only at the optimistic end of one assumption is fragile.
2. **Where is the cost of capital weakest?** The WACC rests on judgment at every input — **beta**, the **equity risk premium**, the **target weights**, the **credit spread** (Chapter 4, Chapter 5, Chapter 6). A small change in any can flip the accept/reject decision, so the estimate's soft spots must be named.
3. **What can the model not capture?** Discounted cash flow ignores **real options** (the value of waiting, expanding, or abandoning), **strategic interactions** (competitor responses), and risks not expressible as a discount-rate bump. These live outside the spreadsheet and must be argued in words.

Example — stress-testing Lumio's production line. The line's +€8 m NPV is modest — about 8% of the €100 m outlay, and a similar fraction of the €108 m of discounted inflows. Two things can erase it. **The cost of capital:** the project breaks even exactly at its 11.5% IRR, so if the true WACC is higher than the 7.8% estimate — the input resting on the most judgment (beta, the equity risk premium) — the cushion shrinks, vanishing at 11.5%. **The cash flows:** a uniform shortfall of about 8% across the four inflows also takes the NPV to zero. The honest recommendation is therefore not “NPV is €8 m, accept,” but “the line creates value provided the cost of capital stays below its 11.5% IRR and the cash flows land within about 8% of forecast — and here is what each assumption is worth.”

Pitfall. A recommendation is defended **not by its point estimate** but by knowing which inputs it depends on and how far it survives their plausible range. Beware the false precision of an NPV quoted to the nearest euro on top of a 6% guessed equity risk premium and a 2% terminal growth rate chosen with no real basis — the apparent rigour of the arithmetic hides the fragility of the assumptions. And beware **advocacy modelling** — a model built to justify a decision already made: a sponsor who wants the project will find a growth rate that makes NPV positive. The critic's job is to surface that assumption and price it.

Exercises

10.1. A firm can invest €120 m now for after-tax free cash flows of €40 m, €45 m, €45 m and €30 m over four years; its WACC is 9%. (a) Compute the NPV and give the invest verdict. (b) The firm is currently under-leveraged; if it funds part of the outlay with new debt, which way does its WACC move, and what does that do to the NPV? (c) After funding this and its other positive-NPV projects it still holds surplus cash and has a stable, income-oriented shareholder base — recommend a payout mechanism. (d) Change one fact: the firm's **full** slate of positive-NPV projects this year needs €300 m, but internal cash is only €90 m and its shareholders rely on the dividend. Explain why it cannot fund the whole slate, hold leverage in its safe range, and protect the dividend all at once, and outline how you would resolve the conflict.

10.2. For the same project, name (a) the assumption the NPV most depends on, (b) the WACC input resting on the most judgment, and (c) one source of value or risk a discounted-cash-flow NPV cannot capture.

Answers. **(1)** (a) $NPV = -120 + 40/1.09 + 45/1.09^2 + 45/1.09^3 + 30/1.09^4 \approx +€10.6 \text{ m} > 0 \rightarrow \text{accept}$. (b) Under-leverage means adding debt **lowers** WACC (more tax shield, still-modest distress risk); a lower discount rate **raises** the NPV, strengthening the case. (c) A steady, modestly growing **dividend** fits stable flows and income-oriented holders; distribute only the cash left after funding every positive-NPV project. (d) The €210 m external gap (300 – 90) is too large to raise as debt without pushing leverage past its safe range, where rising distress risk lifts r_e, r_d and so WACC — eroding the very NPVs; yet cutting the dividend breaks a credible commitment. All three cannot hold at once. Resolve it by ranking the slate by NPV, funding the top projects with the debt that keeps leverage near the optimum, deferring the marginal projects rather than buying WACC-raising leverage, and holding the dividend flat — some positive-NPV euros must wait.

(2) (a) Typically the later cash flows and the discount rate — the €10.6 m cushion is under 10% of the inflows, so a modest shortfall erases it. (b) The cost of capital: beta and the equity risk premium behind the 9% WACC; a rise toward the project's IRR turns NPV negative. (c) Real options (waiting, expanding, abandoning), competitor responses, and risks not expressible as a discount-rate bump.

Further reading

- **A. Damodaran**, *Applied Corporate Finance and Investment Valuation* — the invest / finance / return framework, and the cost of capital treated as a chain of judgments rather than a lookup.
- **J. Berk & P. DeMarzo**, *Corporate Finance* — a thorough reference for the time value of money, NPV and IRR, CAPM, WACC, capital structure, and payout policy.
- **F. Modigliani & M. Miller** (1958), “The Cost of Capital, Corporation Finance and the Theory of Investment” — the original irrelevance proposition that Chapter 7 builds on.
- **W. Sharpe** (1964) and **J. Lintner** (1965) — the papers that introduced the Capital Asset Pricing Model used in Chapter 4.